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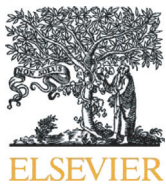
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Exploring Current and Future Situation of Mediterranean Silvopastoral Systems: Case Study in Southern Spain[☆]

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ABSTRACT

Extensive range-based livestock production has suffered a sharp decrease in Mediterranean Spain in recent decades due to two opposing trends in land use: progressive abandonment of practices in marginal rural areas and land use intensification in more productive areas. In this study, we explored the visions of key stakeholders on the current and future situation of silvopastoral systems in the Sierra Morena mountain range of southern Spain, where extensive livestock grazing is declining because of competition with other more profitable activities, such as intensive olive groves and game hunting. We performed a discourse analysis using Q methodology and evaluated the existing relationships of information exchange and conflict among local institutions such as farmer's associations, the provincial council, and conservation organizations using social network analysis. Semistructured interviews were conducted with key stakeholders representing local formal and nonformal institutions involved in extensive livestock production. Results showed three distinct discourses among stakeholders related to profitability, conservation, and modernization. There was a strong consensus on the management strategies that should be promoted to support extensive livestock production and preserve the associated silvopastoral systems. Social networks revealed high cohesion and communication exchange and a low level of conflict among stakeholders, with few exceptions. The low influence of livestock producers on policy decision making largely hinders the development of participatory management schemes. Nevertheless, similarities among the visions and opinions of different institutions suggest that new possibilities might emerge for silvopastoral systems through the collaboration among the different socioeconomic sectors present in the area.

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Introduction

Over the past 200 years, silvopastoral systems in Europe have been extensively modified by different drivers, such as

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demographic and political changes, land ownership changes, and profitability (Jepsen et al., 2015), with direct consequences on land degradation, loss of agrobiodiversity, deterioration of ecosystem services flow, and an increase in trade-offs among land uses and over the management of ecosystem services (Buckles and Rusnak, 2005; Gordon et al., 2010; García Mora et al., 2013; Infante-Amate and González de Molina, 2013a). This situation has increased uncertainty about the long-term sustainability of silvopastoral systems, risking the resilience of these areas, eroding their multifunctionality, and compromising their capacity to provide essential ecosystem services to sustain human well-being (García-Llorente et al., 2012; Nieto-Romero et al., 2014; Salvia and Quaranta, 2015).

Mediterranean silvopastoral systems cover 50% of the surface area of Spain (Pleninger et al., 2015) and are the most extended type of ecosystem in the South of the country (i.e., Autonomous Community of Andalusia). Extensive range-based livestock production has special relevance in Southern Spain, as rangelands have historically had a key role in biodiversity conservation and fire prevention, resulting in

landscapes with what is termed in European policy as having “high nature value” (Gómez-Limón and de Lucio Fernández, 1999; Pinto-Correia and Vos, 2004; Riedel et al., 2013). An example of these landscapes is the *dehesa*, a silvopastoral ecosystem with an overstory of evergreen species of the genus *Quercus* and native herbaceous vegetation with high biodiversity formed as a result of the clearing of shrubland and thinning of woodland areas for extensive livestock grazing. *Dehesas* provide an example of sustainable and traditional land use, considered the most extensive, diverse, and low-intensity land use in Europe, with important socioeconomic value, along with their role in maintaining the traditional ecological knowledge (Gaspar et al., 2009; Carmona et al., 2013). *Dehesas* are also important wintering areas for transhumant livestock and where original breeds of cattle and sheep have been historically reared, including fighting bulls. Transhumance was developed by ancient Mediterranean societies in response to climatic fluctuation, and it is a sustainable practice with a key role in maintaining cultural landscapes and traditional ecological knowledge (Oteros-Rozas et al., 2013). Furthermore, the rearing of bulls for bullfighting is a traditional cultural practice that also has a key role in maintaining autochthonous livestock breeds (Santa-Martina, 2001).

In Mediterranean Spain, silvopastoral systems are suffering from the effects of two opposing drivers of change: 1) the abandonment of less productive extensive production systems in areas that are typically located in rural environments suffering from isolation and accessibility problems (SNEA, 2013) and 2) the intensification of agrarian practices in the most accessible and productive areas. Mechanization, increasing global population, and new dietary preferences associated with higher meat consumption are fostering intensification of livestock production systems (Nonhebel and Kastner, 2011). This phenomenon leads to higher energy consumption and negative impacts on the environment and multifunctionality of silvopastoral systems (Nonhebel and Kastner, 2011; Infante-Amate and González de Molina, 2013b). Since extensive livestock production is an activity with high ecological and socioeconomic value linked to the supply of many ecosystem services, such as meat production, fire prevention, and maintenance of traditional ecological knowledge, it should be protected against intensification and abandonment trends, which may progressively degrade ecosystem service production, and the sustainability of silvopastoral systems (Oteros-Rozas et al., 2014).

Often, some policies or management decisions lead to land use changes that can derive in trade-offs among ecosystem services, which results in changes of the beneficiaries of those ecosystem services, increasing conflicts among stakeholders (Kovács et al., 2015). In addition, solutions to complex environmental problems must consider social, political, and economic dimensions and the multiple interests at stake (Duvall et al., 2016). Thus, it is important to strengthen communication flow and include the perspectives of affected stakeholders in participatory processes for the management of silvopastoral systems and their associated ecosystem services to improve the suitability of decisions for specific situations (Tarrasón et al., 2016). Understanding the social structure and existing relationships within the system may facilitate appropriate interventions oriented toward mobilizing new collaborations and communication flows among different sectors (Prell et al., 2010; Sandström and Rova, 2010). In addition, management plans would be more effective, supportive, and inclusive if different stakeholders were allowed to collaborate in problem identification and decision-making processes (Barry and Proops, 1999). These collaborative approaches also help to realize acceptable situations between human interests and ecosystem health (Duvall et al., 2016). Therefore, understanding the underlying values, perceptions, and motivations of stakeholders is helpful to understanding the role of extensive range-based livestock production within silvopastoral systems, providing a way of identifying which strategies should be adopted to help maintain the sector with no harm to the environment.

Here, we used a case study in Mediterranean Spain to: 1) explore the discourses of different key stakeholders and institutions such as

farmer's associations, the provincial council, and conservation organizations on the present and future role of extensive livestock production (specific objective 1); 2) analyze the influence of each stakeholder group in decision making, as well as potential underlying conflicts among them (specific objective 2); and 3) explore whether social groups with similar opinions had a strong relationship, identifying which ties should be created or reinforced (specific objective 3). We hypothesized that: 1) different discourses regarding extensive livestock production exist in association with a conservation versus development dichotomy; 2) groups sharing similar discourses have a stronger relationship in the social network; and 3) groups holding different discourses have more conflicts. This study is expected to reveal management practices with a high degree of support among stakeholders through Q methodology and to improve governance in a threatened Mediterranean silvopastoral system, through social network analysis combined with Q methodology. Furthermore, this study is expected to provide general insights to help revise current trends in extensive livestock production in the face of global environmental change.

Methods

Study Area

The study area encompasses three watersheds (Rumblar, Guarrizas, and Guadiel Rivers) located in Eastern Sierra Morena (province of Jaén, Andalucía, southern Spain). It contains 14 municipalities and covers nearly 245 800 ha. Four dominant land use types occupy over 60% of the study area: Mediterranean forests and shrublands; intensive olive grove crops; coniferous forest plantations; and oak *dehesas* (Fig. 1). Mediterranean shrublands represent the natural vegetation of the study area, which is mainly composed of holm oaks (*Quercus ilex*), shrubs, and herbaceous vegetation. *Dehesas* are Mediterranean silvopastoral ecosystems that consist of pasturelands with scattered trees, primarily holm oaks (*Q. ilex*), mostly used for extensive cattle ranching. Mixed-pine forests (mainly *Pinus pinaster*, mixed with several other coniferous tree species) are the result of plantation programs developed in the 1960s. Current forestry management practices in these areas combine pine wood extraction, favoring the recolonization of native Mediterranean woody species. Finally, olive groves, managed for olive oil production, represent the most intense land use strategy and the major economic activity.

This area hosts populations of two emblematic and protected species: the Iberian wolf (*Canis lupus signatus*) and the Iberian lynx (*Lynx pardinus*). Sierra Morena hosts the largest existing population of critically endangered Iberian lynx (CMA, 2011). Three natural protected areas cover approximately 5% of the study area: Despeñaperros Natural Park, Andújar Natural Park, and Cascada de la Cimbarra Natural Landscape. In the extensive livestock sector, the existence of various autochthonous livestock breeds is noteworthy, most of which are protected; namely, *montesina* sheep, *blanca andaluza* goat, *negra serrana* goat, *berrenda en colorao* cow, *berrenda en negro* cow, and *pajuna* cow (Fernández Rodríguez et al., 2009).

The main extensive livestock production activities focus on sheep and goats, which represent 33% and 26% of total of livestock heads in the area, respectively, followed by pigs (22%), horses (13%), and cattle (1%), which are mainly raised for meat production and bullfighting (SIMA Andalucía, 2011). Of note, transhumance “on foot” is still practiced along the path linking the Sierra de Albarracín and Sierra Morena through the Conquense Drove Road (Oteros-Rozas et al., 2013). Traditional extensive livestock production modes in the Sierra Morena are progressively disappearing because of the lack of qualified shepherds, poor remuneration of laborers, and limitations in land use, either due to high rents or competition with other activities such as hunting, forestry, and agriculture (EFNCP, 2013; Diputación de Jaén, 2013).

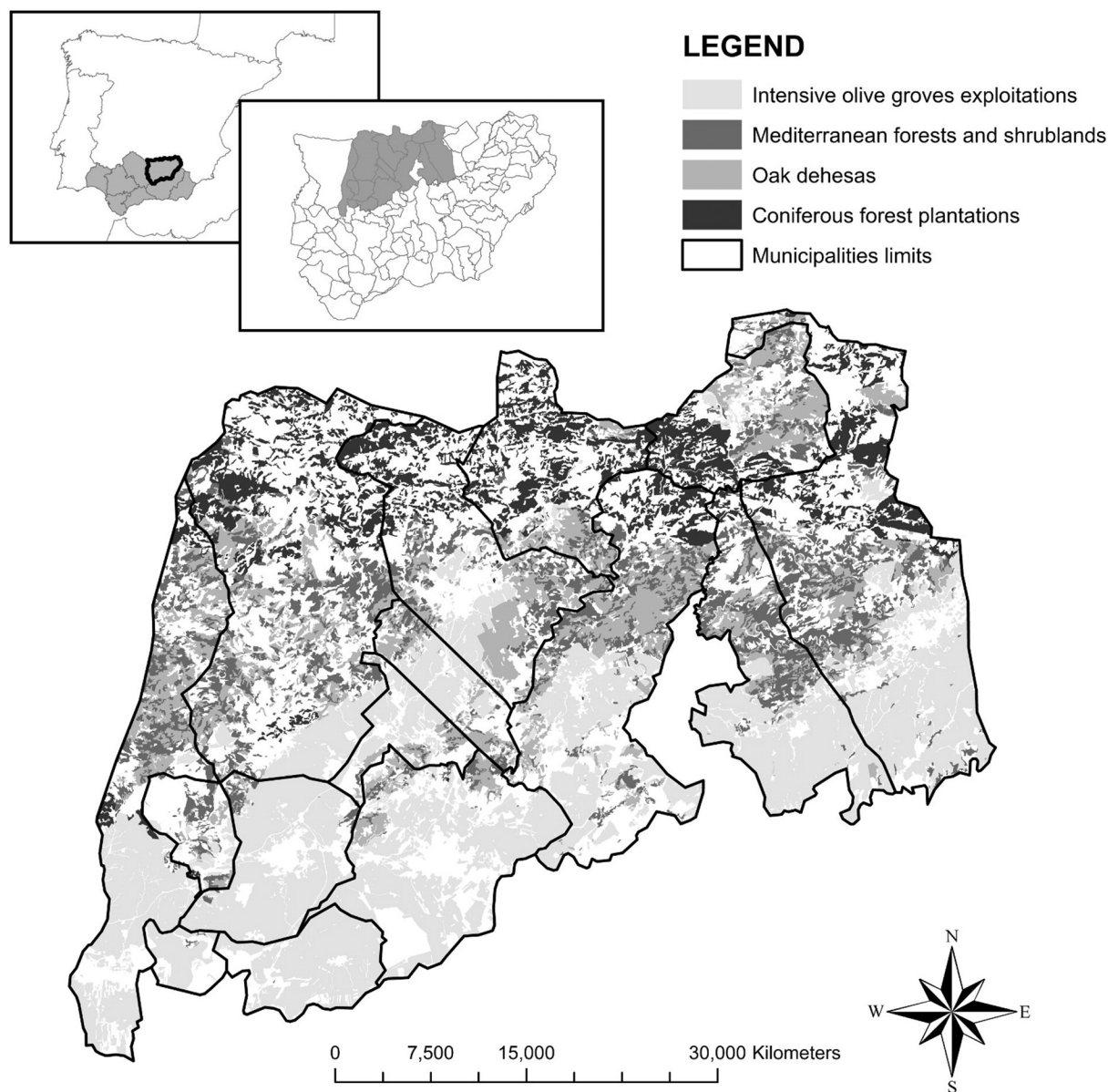


Figure 1. Map of the study area showing the four main land uses in the territory.

In sum, Eastern Sierra Morena is a complex area on which both socioeconomic (hunting, forestry, olive agriculture, etc.) and conservationist (lynx protection, existence of protected areas, etc.) interests act simultaneously and interact with livestock production in different ways. This complexity and the potentially high level of conflict makes the territory of Sierra Morena a suitable study area to analyze how different sectors of the population perceive the role of extensive livestock production. This situation also provides an ideal setting to identify the key stakeholder groups in the area that will potentially influence the development of a productive and sustainable livestock sector.

Methodological Design

The methodological framework proposed to address the specific project objectives includes the use of Q methodology (specific objective 1) and social network analysis (specific objective 2), as well as the combination of the two (specific objective 3). Figure 2 shows the steps of the methodological process that links these two methods. Despite their

common use in social science research, the two methods have seldom been used together to study a particular socioeconomic sector (Easdale, 2015). To perform the two analyses, a series of face-to-face semistructured interviews were conducted during April and May 2015. Interviews also included the collection of general socio-demographic information. All interviews were tape-recorded after obtaining the consent of the interviewees.

Key stakeholders and institutions in the study area were previously identified using the snowball sampling technique (Prell et al., 2010), which began with a list defined by Martínez-Sastre et al. (2017). The interviewees were selected on the basis of the following criteria: 1) they had a formed opinion about extensive livestock production; 2) they represented one or more of all possible interests, perspectives, and experiences within this sector; and 3) they were representative of major formal and nonformal institutions related to rangeland livestock production and land use decision making in the study area (Webler et al., 2001; Bischof, 2010; Zagata, 2010). This approach resulted in contacts with 21 stakeholders (see Appendix A for detailed information).

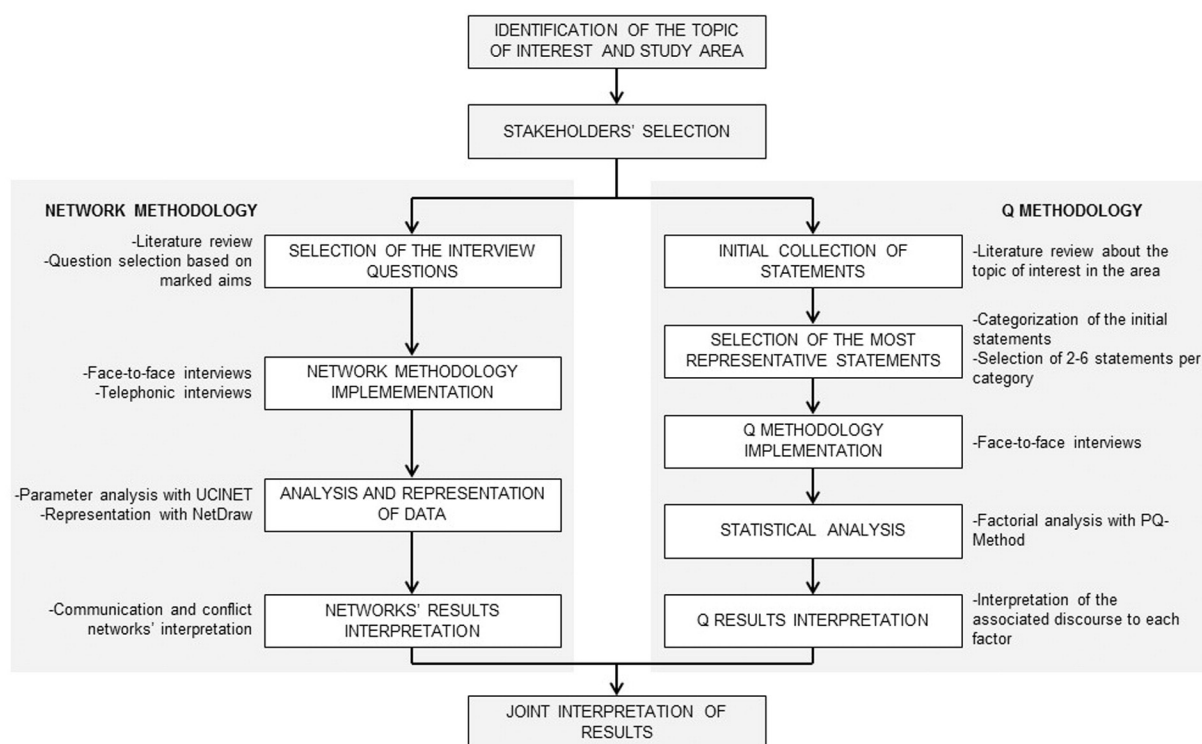


Figure 2. Schematic of the methodological approach, summarizing the major sequential steps followed during the research process.

Q Methodology for Exploring Discourses About Livestock Management

Q methodology was performed to explore the discourses of different key stakeholders and institutions on the present and future role of extensive livestock production, as well as to identify the main potential conflicts among them (specific objective 1). This method was first developed by William Stephenson in 1935 (Durning, 2006), with the aim of revealing the degree of subjectivity of stakeholders in a given situation. Because every human action is motivated by subjective views and perceptions (Duenckmann, 2010), Q methodology is an important tool for identifying the similarities and differences among various stakeholders (Durning, 2006). Moreover, Q methodology analyzes these views with qualitative and quantitative techniques, providing statistical support to qualitative discourses (Kindermann and Gormally, 2013).

We started by collecting statements from a literature review of official policy documents (Natural Protected Areas sustainable development plans and management plans, rural development plans, etc.), and exploring media coverage in regional newspapers reporting about arguments over livestock management in the study area. All of these documents were public, and the interviewees had free access to them (see Appendix B for detailed information). In addition, because this paper is part of a larger research project that started in 2012 in the study area (Martínez-Sastre et al., 2017), the arguments expressed in previous interviews with key stakeholders were also taken into account. Initially, 202 statements were obtained, which were split into eight main classes related to ecological, economical, sociocultural, and management issues, specifically: conflicts, financing, livestock sector situation, participation and training, production/marketing, sustainability and conservation, sustainable livestock practices, and tourism. In a second step, the most relevant and representative statements (between two and six per class) were selected, with the removal of duplicates reducing the final number of statements to 42 (see Appendix C for a complete list of statements).

During the interviews, the 21 selected respondents were asked to answer the following question: “Based on the current situation and existing conflicts affecting the extensive livestock sector in Eastern Sierra

Morena, what strategies related to this activity do you consider most important to prioritize?” Respondents had to rank the statements on a scale from -4 (completely disagree) to $+4$ (completely agree), with statements scoring 0 for those responses considered doubtful or irrelevant (Swedeen, 2006). Sorting was completed in a matrix with a normal distribution that prevents a large number of cells at the extremes and forces the respondents to carefully consider their strongest perceptions (Bischof, 2010). During the Q sort, participants expressed in their own words how they interpreted the statements and why they were organizing them in a certain order. This qualitative information was recorded and later used to obtain a better interpretation of the ranking of statements (Berry et al., 2016).

PQMethod software was used to perform a factorial analysis in which respondents represent the variables (Bischof, 2010; Duenckmann, 2010). This software correlated each Q sort with all remaining sorts and later grouped the respondents into different factors on the basis of their similarities and differences. Thus, stakeholders with similar views formed part of the same factor (López-Gelats et al., 2009; Ellingsen et al., 2010). Following other studies using Q methodology (Barry and Proops, 1999; Kindermann and Gormally, 2013), factors with an eigenvalue > 1 were selected and the accumulated variability explained had to be higher than 60%.

Social Network Analysis for Identifying Key and Conflictive Institutions

Social networks are the set of relationships established among different stakeholders (individuals or institutions). We performed a social network analysis to reveal the influence of each institutional group in decision making, as well as potential underlying conflicts among institutions (specific objective 2). We created two different networks: The first network was used to identify existing communication relationships among institutions, while the second network was used to identify existing conflicts among institutions. The same 21 stakeholders that were selected for Q interviews were also asked to respond to questions related to the social network, serving as representatives of the main institutions that have concerns or that influence extensive livestock

management in the area. This approach allowed us to obtain information about the institutional networks that exist in the study area. During the interview, participants could add other relevant institutions to the communication or conflict network (i.e., free recall). Only two institutions (organic farmer associations and the Provincial Council) were added to the 21 institutions initially identified. When necessary, the original interviews were complemented by telephone communication to incorporate links with these two institutions.

First, interviewees were asked to respond, as representatives of their respective institution, to the following questions about exchange of information:

“From the list of institutions, in the last year, with which ones did you communicate to discuss issues related to livestock management in the area?”

“If this communication existed, how frequently did you communicate with those institutions? (low, medium, or high)”

The answers to these two questions allowed us to identify the communication network within the territory as related to extensive livestock production. In comparison, the conflict network was identified through the answers respondents gave to two questions, which uncovered the presence or absence of conflicts and their intensity:

“With reference to the disputes or confrontations that your institution has had in discussing issues related to livestock management (during the last year), which institutions did you have those with?”

“If conflict existed, what was the intensity of the dispute or confrontation? (low, medium, or high)”

Respondents had to identify the institutions they were in conflict with, selecting them in the same list of institutions that we provided them to complete the communication network.

Once all of the data from the interviews had been collected, a “square” matrix was created and analyzed with the software UCINET (Prell et al., 2009; Downey, 2010; Hirschi, 2010; Prell et al., 2010; Sandström and Rova, 2010). Following Bodin and Crona (2009) and Prell et al. (2009), different parameters of the network were analyzed, specifically tie strength, betweenness, centrality, centralization of the network, and density of the network (see Table 1 for detailed definitions). Networks were also visualized graphically with NetDraw. Finally, groups within

the network were obtained using UCINET command “factions,” which separate the network into different institutional groups on the basis of their similarities (internal group links are maximized and external links are minimized).

Results

Existing Discourses About the Present and Future Role of Extensive Livestock Production

Three significant factors were selected from the Q methodology analysis (Table 2), which explained 66% of the variability. The three discourses advocate extensive livestock maintenance; however, each discourse had a unique perspective. Factor 1 emphasized the profitability of the livestock sector; factor 2 had a more conservationist dimension; while factor 3 considered sector modernisation to be necessary. Hereafter, we present the three discourses that emerged from the way in which the statements were ranked by participants (statement number in parenthesis) (Table 3) and highlight some quotations from the interviews associated with each particular discourse.

The first factor (explaining 54% of variability) was associated with a discourse of “Enhancing traditional livestock profitability through the empowerment of farmers and environmental coalitions.” This discourse was supported by nine participants, mainly from institutions related to administration and public organizations (see Table 2). This factor shows support for extensive livestock and transhumance maintenance in Sierra Morena (statement [S] 36), both to preserve genetic diversity (S31) and to avoid the disappearance of this traditional practice, which produces high-quality food products (S41).

“Transhumance needs to be improved and the sacrifice and hard work behind this practice must be reduced. This practice should not be abandoned because livestock grazing is necessary.”

“Native cattle breeds are less productive, but they are highly adapted to their environment, so they could make an area profitable that would, otherwise, be unproductive.”

Strategies for keeping this practice were related to the association and empowerment of farmers (S24), facilitating an increase in

Table 1

Definition and utility of the network methodology parameters considered in the project. (Adapted from Bodin and Crona, 2009 and Prell et al., 2009).

Parameters	Definition	Utility
Ties' parameters		
Strong ties	Strong relationships between the nodes	Communication network: high communication frequency and high influence between nodes, easier exchange of complex information, but this information is frequently redundant Conflict network: high conflict degree between the nodes
Weak ties	Weak relationships between the nodes	Communication network: low communication frequency within the network; higher resilience and ease of including new ideas within the network; low level of trust between the nodes; harder exchange of complex information Conflict network: low conflict degree between the nodes
Nodes' parameters		
Betweenness degree	Degree in which a node connects other 2 nodes that could not otherwise be connected	Information about the power and information access of the nodes, as well as the easiness for incorporate and manage new ideas to the network Information about the facilitation of information transmission and the freedom of interaction among network components
Centrality degree	Indegree Number of ties that other nodes have with a given node	
	Outdegree Number of existing ties between a node and the others	
Network parameters		
Centralization degree	If it has a high value, it means that a few individuals hold most of the links of the network	Information about the ease of solving problems and coordinating the nodes of the network
Density	Result of dividing the number of existent connections between the number of all possible connections for a node	Information about the confidence, reciprocity and homogeneity degree, as well as the understanding facility within the network

Table 2

Three factor solutions selected for the Q method analysis. Eigenvalues, explained variability in percentage, number of actors, and main sectors associated with each factor are presented.

	F1	F2	F3
Eigenvalue	11.26	1.42	1.17
% Expl. variability	54.00	7.00	6.00
% Expl. vs. accumulated	54.00	61.00	67.00
Number of actors associated with each factor	9	7	3
Institutions associated with each factor	DAO ^a , WEA, NPA Administration, Provincial Council, Environmental NGO, tourism companies, RDA, forest rangers, researchers	Andalusian Association in Defence of Transhumance, fighting bull producers, Nueva Mesta de Albarracín Association, FDA, private game producers, LHPA, autochthonous livestock breed associations	Agricultural unions, livestock dealers, olive production associations

^a All institutions abbreviations are spelled out in Appendix A.

profitability. Following this line of argument, the major beneficiaries of the Common Agricultural Policy (CAP) should be the farmers, to the detriment of the land owners (S18), while intermediate dealers in the sale of livestock should disappear. This approach would help farmers to obtain higher economic benefits (S23). All of these changes, added to a new production model that integrates meat and oil production within the same area (S17), would create a new situation in which farmers collaborate with one another to obtain more power on the decisions that affect their own activities. In parallel, this approach would allow the maximum possible production to be generated from the territory, increasing the profitability of the practice and its maintenance.

“The livestock farmers association should be fostered if producers really want it, but someone must give them the necessary tools to accomplish this [...] we all need a helping hand for team work, otherwise collective projects will not work out.”

Within this factor, a clear tendency toward conservation also emerged, in which the conservation of wolves was defended (S6). The opinions behind this discourse were consistently linked with supporting the introduction of livestock to forests and protected areas, because this activity (if properly managed) is perceived as favorable for ecosystems conservation. Therefore, the stakeholder groups within this factor strongly defended livestock and environmental conservation but stressed the economic component, mostly referencing new ways of commercialization and production.

The second factor (7% of explained variability) was associated with a discourse focused on *“Enhancing livestock multifunctionality and product quality.”* This discourse was clearly in favor of all aspects of livestock maintenance (S12), from transhumance (S36) to the rearing of fighting bulls. Rearing bulls is perceived as having important economic potential, due to it being a traditional practice with important cultural and natural values. This discourse is associated with institutions directly linked to the extensive livestock sector (see Table 2).

“Transhumance must be promoted, because it provides many services, such as seed dispersion [...]. Livestock drove roads are more important than highways.”

In addition to the desire to maintain this activity because of its intrinsic values, respondents felt it is also important to maximize economic outputs. Low meat prices are not desirable because livestock meat is a high-quality product that can compete in the market (S11).

“If meat prices were fair, there would be no need to give subsidies to the livestock producers.”

However, defending livestock does not mean that productivity is the only focus; livestock were also considered important for environmental conservation, contributing to fire prevention (S14). Finally, most members of this group were concerned about the risk of disease transmission

Table 3

Statements significantly associated with each factor (we only show statements with relatively high or low rankings to simplify the table; Appendix D presents the complete table). N is the number given to each statement. Rank is the average of that given by the members of the factor to each statement.

N°	Statement	Rank	Score
Factor 1			
24	Associations among local shepherds should be promoted to improve production and product commercialization processes.	4	1.63**
17	Integrated production of meat and oil within the same area should be promoted.	4	1.39**
41	Livestock products derived from transhumance should be promoted because they have higher quality and added value compared with other products from the area.	4	1.39**
31	The rearing of traditional livestock breeds must be supported to maintain genetic diversity.	2	0.84**
18	CAP grants should mainly be given to land owners, not to farmers.	−2	−1.16*
23	Marketing should be made by dealers to maximize efficiency.	−2	−1.23*
36	The abandonment of transhumance must be promoted, because of the huge sacrifice and work that it requires.	−4	−1.95*
6	Wolves must be hunted because they damage livestock.	−4	−1.96**
Factor 2			
14	Livestock should be introduced to areas vulnerable to fire to reduce the risk of fire.	4	1.25*
28	The possibility of disease transmission between livestock and wildlife should be avoided.	3	1.24**
12	Livestock production should not be fostered because it has reduced economic value; it is more profitable to maximize other activities, like hunting.	−2	−0.94**
36	The abandonment of transhumance must be promoted, because of the huge sacrifice and work that it requires.	−3	−1.45*
11	Locally produced meat prices should be lower to compete with other zones.	−4	−1.59**
Factor 3			
14	Livestock should be introduced to areas vulnerable to fire to reduce the risk of fire.	4	2.04*
20	Livestock activity should be controlled to avoid negatively impacting the conservation of mountain regions in the Mediterranean.	3	1.44**
5	Native cattle breeds are less productive, so they should not be fostered.	3	1.32**
3	The area dedicated to livestock must be increased because olive groves have led to the loss of area dedicated to this activity.	2	0.65*
39	Tourism and the culture linked to the fighting bull sector should be fostered.	−2	−0.79*
21	Communication campaigns and environmental education for shepherds should be promoted to minimize negative impacts to Iberian lynx.	−2	−1.19*

* Significance $P < 0.05$.

** Significance $P < 0.01$.

between livestock and wildlife (S28). These stakeholders believe that the infection of livestock with wildlife diseases, such as tuberculosis, is frequently caused by animals drinking from the same drinking troughs. Consequently, the members of this group believe that it is important to implement health control programs for wildlife in hunting areas. This action would prevent the expansion of disease and its associated economic damage.

“Disease transmission between livestock and wildlife is a problem and it is expensive for the producers [...] You cannot conduct livestock and hunting production simultaneously in these conditions, so you have to construct fences, control the drinking troughs, etc.”

Finally, the third factor (6% of explained variability) was associated with a discourse focused on “promoting livestock modernization.” This was the only discourse pointing to the abandonment of some traditional livestock practices. For instance, native cattle breeds are perceived as not providing enough benefits (S5), while the empowerment of other activities, such as rural tourism associated with livestock, was disregarded (S39).

“I know fifty dehesas and only one has accommodation for rural tourism. It would be nice to foster tourism associated with livestock grazing, but it's not profitable.”

This group was made up of three institutions that were mainly related to the livestock sector (see Table 2). In this discourse, livestock were considered important in helping with fire prevention (S14), as long as livestock management is respectful of the environment (S20).

“The best way to prevent fires is livestock grazing, as long as the introduction of livestock is managed properly [...]. There have been fires that suffocated themselves when they got to a grazed area.”

The opinions behind this discourse were not advocating for the removal of livestock, but they did advocate the removal of customary practices considered too demanding in relation with the benefits received.

“Transhumance derived products are not profitable, so their commercialization should not be fostered.”

Finally, the lowest-ranked statement in this discourse is related to communication campaigns and environmental education to minimize harm to the Iberian lynx and its habitat (S21), as producers do not harm this animal because it does not attack livestock, with very few exceptions when lambs are attacked.

Influential and Conflictive Groups Among Extensive Livestock Production Institutions

Our findings for the exchange communication network show that there is high cohesion, along with a good degree of understanding among the members of the network (density of 80.4%) (Table 4). It is noteworthy that the exchange of information (regarding the outdegree, indegree, and betweenness parameters) among institutions was two-way (i.e., institutions that often communicate with other institutions also receive more information from the other institutions). Furthermore, a group of institutions with high communication capacity frequently helped to link groups that would not be connected otherwise. This situation was sustained by institutions with high values for those three network parameters (e.g., LHPA¹, DAO, forest rangers, and the local administration). In contrast, institutions with low values for these parameters did not have sufficient communication capacity to help in linking other groups (e.g., researchers, olive production cooperatives, livestock dealers) (see Table 4 and Fig. 3A).

The situation of livestock producer organizations is noteworthy. Our results showed that transhumant shepherd's associations and autochthonous livestock breed associations lack power and influence within the community. Their low capacity to connect institutions that otherwise would not be connected shows the difficulty of incorporating and managing new concepts into the network. Our results only identified the breeders of fighting bulls as a powerful livestock production institution (with one of the highest degrees of betweenness). This finding matches the opinions from different sectors during interviews that this group has a strong influence on decision making about livestock management. The most powerful institutions (i.e., betweenness degree higher than 20) mainly belonged to the technical and administrative sector (LHDA, Provincial Council, DAO, local municipalities, and forest rangers), confirming the unequal distribution of power within the network (see Table 4).

Our results from the conflict network show that few conflicts exist within the network (density of 18.2%) (Fig. 3B). For instance, some institutions such as the Andalusian Association in Defense of Transhumance and the agricultural unions identified some conflicts with other institutions that were not identified by the institutions they said they were in conflict with (i.e., these institutions had higher outdegree values compared with their indegree values). In comparison, some institutions (such as environmental NGOs, private game ground owners, DAO, bull-fighting bull rearers, LHPA, and livestock dealers) had more conflicting situations than they realized. These institutions identified fewer conflicting situations with the other institutions than the other institutions identified with them (i.e., their indegree values were higher than their outdegree values) (see Table 4).

When comparing the results of the communication and conflict networks, institutions that had the potential to link other institutions that, otherwise, would not be connected (i.e., with high betweenness in the communication network) were also the institutions that had a higher number of conflicts (i.e., betweenness higher than 20 in the conflict network). These institutions were mostly those with decision-making capacities, such as the Provincial Council, local municipalities, and NPA Administration.

Relationships Between Institutions Holding Similar Discourses

Combining Q methodology and social network analysis allowed us to explore whether social groups with similar opinions had strong relationships and to identify which ties should be created or reinforced (specific objective 3). The two networks were split into three different groups of institutions based on their similarities. This approach allowed us to compare these network groups with the groups emerging from the discourse analysis (Q factors).

Regarding the communication network, the first group was formed by 10 institutions, half of which belonged to administrative entities or public agencies, while the other half belonged to other sectors. The second group was formed by eight institutions that were livestock associations and rancher groups. The third group was formed by three institutions, one related to the livestock sector and another two related to other activities. When comparing the institutions linked to these communication groups with the institutions linked to each Q factor, we found that 1) 77.8% of the institutions within Q factor 1 were also linked to UCINET communication group 1; 2) 71.4% of the institutions within Q factor 2 were also linked to UCINET communication group 2; and 3) none of the institutions within Q factor 3 were also linked to UCINET communication group 3 (Fig. 4). In the conflict network, the first group was formed by nine institutions, five administration or public organizations, three related to other activities, and one livestock association. The second group was formed by six institutions, four related to the livestock sector and two related to tourism and rural development. The third group was formed by six institutions, four related to the livestock sector and two related to other associations. When comparing the institutions linked to these conflict groups with the institutions linked

¹ All institutions abbreviations are spelled out in Appendix A.

Table 4

Values of density, centrality, betweenness, and centralization for each of the institutions included in communication and conflict network. Sector legend: 1) associations related to livestock sector; 2) administration and public organisms; 3) livestock farmers; and (4) other socioeconomic activities.

Sector	Institution	Communication network			Conflict network		
		Density	80.4%	Betweenness	Density	18.2%	Betweenness
		Centrality			Centrality		
		Outdeg.	Indeg.		Outdeg.	Indeg.	
1	LHPA	29	23	31.4	0	3	0.0
1	DAO	28	23	28.2	2	8	9.1
2	WEA	27	24	11.7	8	4	27.4
1	Forest rangers	26	29	20.9	5	4	19.4
2	Local municipalities	26	28	27.7	9	9	63.3
2	NPA Administration	26	20	14.9	10	8	29.2
1	FDA	22	12	9.5	6	3	2.5
3	Nueva Mesta de Albarracín Association	21	20	12.5	1	0	0.0
1	Agricultural unions	21	7	8.2	8	1	7.3
2	PAA	20	29	28.2	5	9	30.4
3	Andalusian Association in Defence of Transhumance	20	16	10.4	10	0	0.0
3	Fighting bull producers	20	14	21.9	0	4	0.0
4	Environmental NGO	17	16	6.4	4	11	21.8
4	RDA	17	14	7.9	3	1	3.0
4	Tourism companies	16	15	6.6	5	0	0.0
3	Autochthonous livestock breed associations	15	18	7.4	6	1	1.0
4	Hunters' societies	15	16	6.0	8	6	40.8
4	Private game producers	14	18	4.7	0	6	0.0
3	Livestock dealers	14	9	2.6	0	3	0.0
4	Olive production cooperatives	7	14	2.4	2	2	7.8
4	Researchers	6	12	6.5	0	0	0.0
2	Provincial Council ^a	0	23	0.0	0	9	0.0
3	Organic farmers associations ^a	0	7	0.0	0	0	0.0
	Centralization	53.7%	53.7%		28.5%	33.3%	

^a Institutions that could not be reached for the interviews, hence their rank of zero for both their outdegree and indegree.

to each Q factor, we found that 1) 55.6% of the institutions within Q factor 1 were also linked to UCINET conflict group 1; 2) 42.9% of the institutions within Q factor 2 were also linked to UCINET conflict group 2; and 3) 66.7% of the institutions within Q factor 3 were also linked to UCINET conflict group 3 (see Fig. 4).

Discussion

This empirical study provides two main contributions for improving our understanding on the status of extensive livestock production in Mediterranean Spain. First, this study described differences in opinions about present and future options for extensive livestock management and the associated silvopastoral systems. Second, this study identified potential alliances that could be reinforced within the community, as well as those social conflicts that need mediation. Finally, this study contributes to a thorough discussion on the limitations and usefulness of combining Q methodology and social network analysis in strengthening the understanding of the functioning and structure of the system.

Converging Discourses and Social Alliances in the Livestock Management Sector

The three factors resulting from Q methodology implementation were highly correlated, particularly when contrasted to other studies (Bredin et al., 2015). Factor 1 explained over 50% of variability, with an eigenvalue of 11.26, which is very high compared with other studies (Pelletier et al., 2000; Nijnik and Mather, 2008; Di Masso and Zografos, 2015). This result confirmed the existence of high cohesion, complementarity, and degree of understanding of the role of traditional livestock production. This activity is considered compatible with all other land uses; that is, silvopastoral systems have traditionally been seen as multifunctional landscapes where different socioeconomic activities and natural dynamics have shaped the landscape (Hadjigeorgiou et al., 2005; Nieto-Romero et al., 2014). In most interviews, participants clarified their visions of the issue, claiming that, within the study area, there is enough territory to implement the different existing

socioeconomic activities (olive groves, conservation, hunting, tourism). They believed that it is not appropriate to maximize any of these activities to the detriment of the other activities. Moreover, in some cases, livestock were seen as helping to improve other socioeconomic activities. For example, in the case of organic olive groves, sheep could help fertilize the soil (Sánchez Rodríguez et al., 2011). Furthermore, the introduction of livestock to coniferous forests or Mediterranean shrublands and rangelands is very important for fire prevention, as well as fertilization and seed dispersion (Gómez Sal, 2007). Finally, extensive livestock production was considered, in some cases, as a potential resource for agrotourism (a sector that is increasing and, in some cases, improving livestock profitability; Leco et al., 2013) because some visitors enjoy views that include livestock grazing on the landscape when visiting *dehesas* and farms or participating in transhumance on foot). This activity also creates opportunities for the development of environmental education and the maintenance of traditional ecological knowledge (Wolf et al., 2016). This land use compatibility differs from the trade-off situation reported by previous studies, where management choices promoted the optimization of certain land use types, leading to the reduction of others (Martín-López et al., 2014).

This cohesion and consensus in the views of different respondents were also supported by social network methodology results. First, the high density observed in the communication network (Prell et al., 2009; Hirschi, 2010; Sandström and Rova, 2010); its centrality (which fell within normal ranges; Hirschi, 2010; Sandström and Rova, 2010); and the equilibrium between strong and weak ties may have been related to the existence of trust among all parts of the system, facility for conflict resolution, and coordination among all the parts in the network (Bodin and Crona, 2009). These results may also be related to the ability to transmit complex information, along with network resilience (Prell et al., 2009).

The social network results also allowed the system to be analyzed to identify key, outcast, or mediator institutions. The communication network results showed that the livestock sector had an alarmingly low capacity to link institutions that would, otherwise, not be connected (betweenness degree). This issue might indicate an important lack of

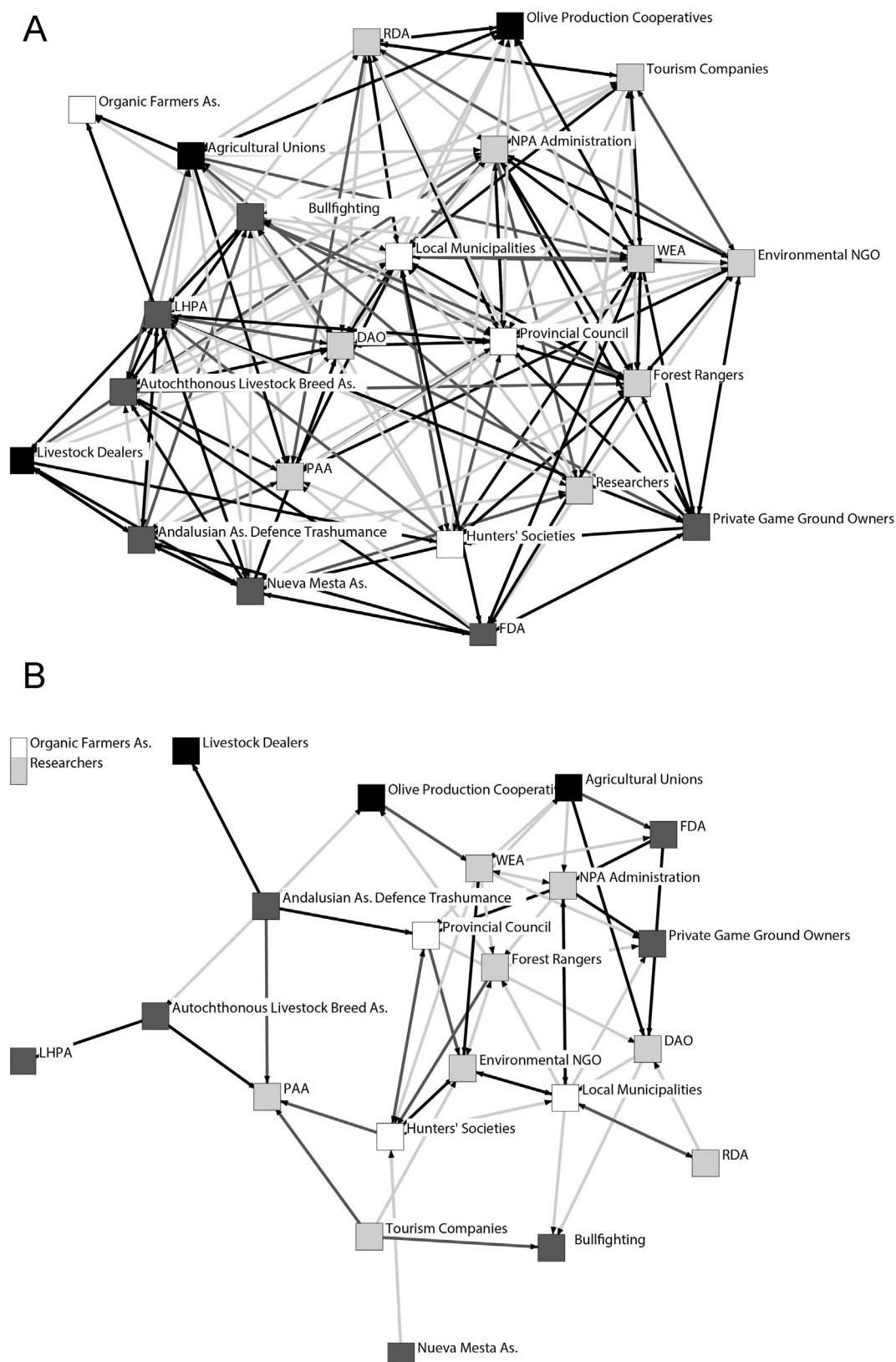


Figure 3. **A**, The communication network. Tie colors represent the communication frequency between the nodes, with light gray indicating a frequency of 1, dark gray a frequency of 2, and black a frequency of 3. Node colors represent the factors the institutions belong to: Light gray is the members of factor 1, dark gray the members of factor 2, black the members of factor 3, and white those that don't belong to any factor. **B**, Conflict network. Tie colors represent the conflict intensity between the nodes, with light gray indicating an intensity of 1, dark gray an intensity of 2, and black an intensity of 3. Node colors represent the factors the institutions belong to: Light gray is the members of factor 1, dark gray the members of factor 2, black the members of factor 3, and white those that do not belong to any factor. Figures made with *NetDraw* software.

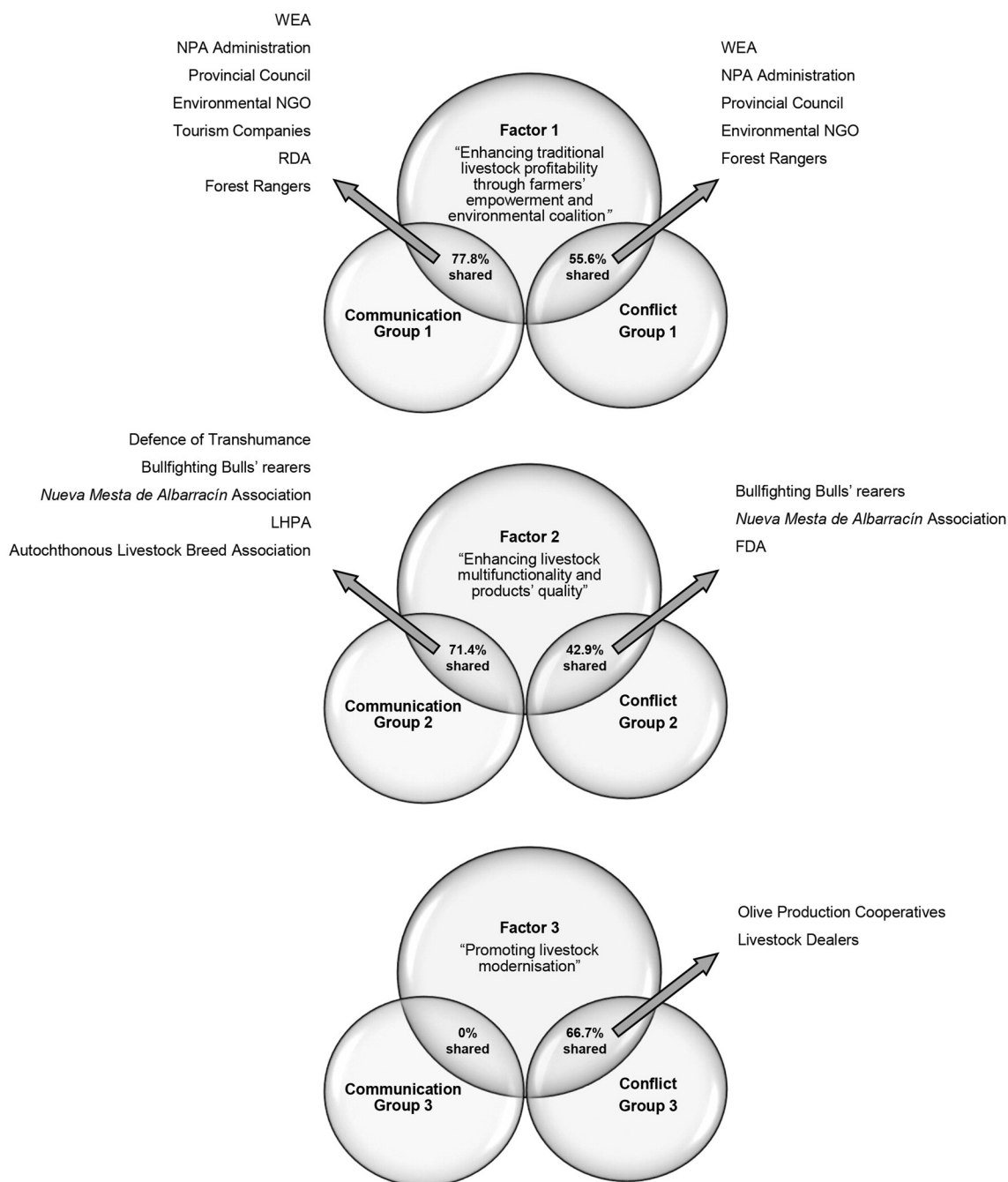


Figure 4. Percentage of members shared by each Q factor and their respective UCINET communication and conflict groups. The numbers inside the overlapping area between circles show the percentage of institutions of each Q factor that also belong to its respective communication or conflict group, and the arrows point to the list of institutions shared by both groups.

power (Prell et al., 2009) in relation to decision making affecting the livestock sector. During the interviews, many of the participants thought that this situation needed to change immediately, with it being repeatedly argued that the association and empowerment of the producers should be improved. This claim has also been common in previous studies that have tried to visualize the future of pastoralism in Spain (Oteros-Rozas et al., 2013). Of note, researchers held very low centrality and betweenness in the communication network. This issue highlights the limited role that researchers currently have in the study area, as well as their lack of power. However, during the interviews, most participants claimed that the influence and interactions of researchers with the community should be strengthened.

Although some institutions show a clear lack of power, the conflict network results show that there are few conflicts among institutions and that the existing conflicts generally have low intensity. Three interesting conclusions were obtained when analyzing the conflict network results. First, there was noticeably high discomfort with environmental associations, which is striking when taking into account that respondents showed a general interest in environmental conservation during the interviews. This phenomenon may be because 1) the institutions are not taking the necessary measures to preserve the environment with the support and collaboration of citizens or 2) the institutions are implementing the right measures, but they are not correctly informing the rest of the population about these actions.

Cases where stakeholders had different perceptions about a conflict (high difference between outdegree and indegree values within the conflict network) were also of interest. Such cases may exist because 1) the representatives of the institutions answered the interviews from a personal viewpoint, focusing on their own conflicts with particular persons in the other institution; 2) the perception of conflict differs among participants, and the conflicts were viewed as more serious or intense by one of the institutions; or 3) some interviewees might not be confident enough to talk about the importance of conflicts with other institutions. The conflict network also helped to identify institutions that may act as catalysts in conflict resolution through conciliation or mediation between participating institutions. Conciliation requires a neutral third party to communicate separately with the institutions in conflict to minimize tension and to find an agreement between them; therefore, within the network, the conciliator must be a node that has the same relationship with both institutions in conflict (Buckles and Rusnak, 2005). In contrast, mediation is the inclusion of a neutral third party that has no power over either conflicting party. Within the network, the mediator is not directly linked with either institution in conflict (Buckles and Rusnak, 2005). Institutions with more serious conflicts (intensity value of 3) are shown in Table 5, along with those who could act as conciliators or mediators. Regarding conflict conciliation or mediation, institutions linked to the administrative sector were the most conflictive groups, but they were also identified as “key stakeholder groups,” because they have high centrality and/or betweenness degrees (Prell et al., 2009). Thus, these groups have the capacity to influence others with their discourse. In addition, such groups would be those considered opinion leaders, which have a strong impact within the community to transform a given situation (Bodin et al., 2006; Crona and Bodin, 2010). This could place the development of collaborative processes at risk, if conflicts are not managed from the outset.

Finally, most stakeholders that had a similar discourse (i.e., factors 1 and 2) had good communication among them (i.e., communication network groups 1 and 2). Thus, establishing and maintaining new

communication bridges between nonconnected institutions might help achieve a better understanding between parties, as well as improving the collaborative processes within society.

Strengths and Limitations of Combining Q Methodology and Social Networks Methodologies

Several authors have analyzed the usefulness of implementing Q methodology and social network analyses. Q methodology analysis is used to reveal the opinions and arguments of a given situation (Durning, 2006) and shows how a given population understands a given phenomenon based on their perceptions and values (Zagata, 2010). Social network analysis provides useful information about the set of relationships established among different stakeholders (individuals or institutions), their interactions and structure, and how information and resources move among them (Spielman et al., 2011). The main methodological contribution of our research is the combination of both methods. We argue that this combination is pivotal, as a first step, to understanding how to improve participation within the community and to understanding the structure of the social system related to extensive livestock production (e.g., main stakeholders and their interactions) and how stakeholder agencies are acting (e.g., the main motivations and explanations of behaviors underlying the way the system is functioning).

The combined use of Q methodology and social network analyses helps to improve our understanding about the structure and functioning of complex systems. For instance, Q methodology objectively quantifies the different perspectives within the community, as well as the differences that exist among stakeholder groups (Kindermann and Gormally, 2013). Social network analysis provides information about the position of stakeholders in the system and their relationships. Combining the two approaches facilitates our understanding of convergences and divergences among stakeholders, as well as potential conflicting issues and the reasons for these conflicts. Such insights

Table 5

Conflicts in the area and the institutions involved. For each conflict, institutions that could act as possible conciliators or mediators are listed on the basis of strong relationships with the parties in conflict (i.e., communication frequency of 3). The potential conciliators are those institutions that have the same relationships with both parties in conflict (in this case we selected those with the same communication frequency with both parties). The potential mediators are those neutral institutions that are not directly linked to any of the parties in conflict.

Institutions involved in main conflicts	Potential conciliator	Potential mediator
Livestock dealers—Andalusian Association in Defense of Transhumance	LHPA, Nueva Mesta de Albarracín As.	NPA Administration, Organic Farmers Assoc. ^a , tourism companies, RDA, environmental NGO, private game producers, agricultural unions, WEA
LHPA—Autochthonous Livestock Breed Assoc.	Nueva Mesta de Albarracín As., Fighting Bull Producers	Environmental NGO, olive production cooperatives, WEA
Autochthonous Livestock Breed Assoc.—Provincial Council	-	Nueva Mesta de Albarracín Assoc., private game producers, olive production cooperatives, researchers
Andalusian Association in Defense of Transhumance—PAA	LHPA	Organic Farmers Association
PAA-NPA Administration	-	Organic Farmers Assoc., livestock dealers
NPA Administration—FDA	-	Organic Farmers Assoc., livestock dealers, hunting societies, olive production cooperatives
FDA-DAO	Autochthonous Livestock Breed Assoc., Nueva Mesta de Albarracín As.	Organic Farmers Assoc., tourism companies, livestock dealers, Environmental NGO, researchers
DAO—Agricultural Unions	—	Tourism companies, forest rangers
NPA Administration—Local municipalities	—	Andalusian Association in Defense of Transhumance, Nueva Mesta de Albarracín Assoc., Organic Farmers Association
Local municipalities—environmental NGO	Provincial Council	Andalusian Association in Defense of Transhumance, Organic Farmers Assoc., Nueva Mesta de Albarracín Assoc.
Environmental NGO—hunting societies	Private game producers	Organic Farmers Assoc., Andalusian Association in Defense of Transhumance, Autochthonous Livestock Breed Assoc., agricultural unions, FDA
Environmental NGO—WEA	-	Andalusian Association in Defense of Transhumance, Organic Farmers Assoc., Nueva Mesta de Albarracín Assoc., Autochthonous Livestock Breed Assoc., LHPA
NPA Administration—private game ground owners	Environmental NGO	Andalusian Association in Defense of Transhumance, Nueva Mesta de Albarracín Assoc., Organic Farmers Assoc., livestock dealers, olive production cooperatives, Provincial Council

^a The Organic Farmers Association is a potential conflict mediator in almost all cases. As it is identified as practically nonexistent in the area, it would not be a good option for conflict resolution.

could help toward generating trust among institutions and increasing cooperation among stakeholders, which, in turn, would improve their capacity to deal with environmental changes. In fact, our findings show that institutions with stronger communication links have similar motivations and convergent interests with respect to the future adaptive management of silvopastoral systems. Overall, establishing new communication bridges among unconnected institutions could generate a better understanding and cooperation within the territory. Moreover, this approach could improve the development of future participatory processes because the application of both methodologies provides an initial step in two-way communication and co-learning processes among researchers and local stakeholders.

With respect to decision making and territorial management, we argue that the results obtained using both methods could facilitate the prioritization and establishment of new policies. The Q methodology results cannot be generalized to the entire population (Dziopa and Ahern, 2011; Pereira et al., 2016; Winkler and Nicholas, 2016) due to the limited number of participants. This issue hinders the identification of all existing discourses within the territory (Reed et al., 2009). However, Q methodology provides a robust analysis of existing beliefs and opinions (López-i-Gelats et al., 2009; Zanolli et al., 2015; Pereira et al., 2016). When we combined this type of analysis with the identification of the main institutions related to the extensive livestock sector and their position in the social network, we obtained enough information to prioritize the main issues of concern. Of further importance, we were able to identify the institutions that could address these issues. This knowledge is very helpful for developing new policies and for decision-making processes.

Combining Q methodology and social network analysis allows the identification and resolution of conflicts through identifying the intensity of conflicts, the components of a given conflict, and the potential mediators or conciliators (Buckles and Rusnak, 2005; Prell et al., 2010; Rathwell and Peterson, 2012). In addition, the combination of these two analyses allows the motives behind such conflicts and the main conflict themes to be elucidated. Therefore, both methods provide an understanding that might help obtain consensus for actions across all institutions, as well as reveal the underlying conflicts in the community in a realistic way. Such information would help overcome obstacles to productive negotiations. This holistic understanding of the explicit and latent conflicts may facilitate their management.

Some limitations also emerged when using Q and social network methods together that should be acknowledged, with the main limitation being the selection of stakeholders. The social network analysis would have been much more rigorous if we had interviewed more stakeholders, but the Q methodology requires that the number of respondents represent about half of the selected statements (Dziopa and Ahern, 2011; Bischof, 2010; Mazur and Asah, 2013), which limits the number of potential interviewees. We included all of the institutions related to the extensive livestock sector, but we only selected one representative stakeholder for each institution so that we could rigorously compare the Q methodology and social network analysis results. This approach largely simplified the social structure but also made it easier to analyze the social network (Marshall and Staeheli, 2015) and to compare both methodologies.

Another potential limitation is the fact that some stakeholders might not have answered as representative of their institutions, leading to a bias in our research, as their institutional view could be different from their personal views (Primmer et al., in press). This limitation could be avoided by performing a second social network analysis focusing on the relationships among individual interviewees. This approach would allow respondents to distinguish institutional and personal points of view.

Finally, we should acknowledge that different ontologies lie behind the two methods. While Q methodology focuses on the subjectivity of stakeholders, the social network analysis focuses on the institutional structure within the study area. However, such ontological discrepancy could be considered a strength because the different approaches to the

subject of study complement each other. Indeed, the social network analysis shows the social structure of the system, while Q methodology completes the network with the underlying subjectivities and motivations of the stakeholders who hold it. Further research may improve the work proposed by adopting a clearer hybridization of both methods (i.e., the development of “discourse networks”).

Regardless of these limitations, the combination of both methodologies has turned out to be a powerful tool for studying a complex system, identifying conflicts, fostering participation and conflict resolution, developing territorial policies, and facilitating decision making.

Implications

Mediterranean silvopastoral systems have suffered the consequences of two different and opposite drivers of change: abandonment of marginal rural areas and intensification of the most productive areas. To achieve the socioeconomic sustainability of these systems, we must identify and understand the different discourses, relationships, and conflicts that exist among the institutions that share the landscape. This approach would facilitate social participation in identifying strategies to improve livestock management practices and the future sustainability of silvopastoral systems. In fact, providing an in-depth overview of the social complexity of the livestock sector allows priority issues to be identified by stakeholder groups seeking a more sustainable and promising future. The combined results of Q methodology and social network analysis highlighted the existence of strong consensus and trust among different stakeholders, which helps overcome obstacles more easily. However, our results also demonstrate the existence of several threats to the extensive livestock sector. Some of these threats include 1) conflicts of interest with, and the pressure from, other, more productive, land uses (i.e., hunting, olive groves, etc.); 2) the lack of power of livestock farmers in territorial management and decision making; and 3) low participation among institutions related to the extensive livestock sector.

Therefore, some management strategies should be reevaluated to achieve sustainability in the sector. Our results support the need to 1) improve commercialization processes through producer's associations, 2) foster new production strategies with higher profitability (i.e., integrated production of meat and oil within the same area), and 3) empower producers to have greater influence on the pricing and commercialization of products to avoid the excessive intervention of dealers. The development and fostering of new public interventions and policies is needed to realize this goal. New policies should address the following issues: 1) resolving or mediating existing conflicts, particularly those involving the most powerful institutions; 2) promoting the empowerment of livestock farmers in decision making; and 3) increasing participatory processes to facilitate the creation and establishment of new actions accepted by all the institutions and groups. The implementation of such actions is expected to help improve the management of silvopastoral systems and the sustainability of the extensive livestock sector.

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